A COMPUTER-ASSISTED RETAINED-SPINE BOUND FOR DIAGONAL RAMSEY NUMBERS

CHENG XIN

ABSTRACT. Conditional on the version-pinned Yang–Mao retained-spine and book interfaces stated explicitly below, we give a source-relative computer-assisted proof of

$$R(k,k) \leq (3.780685290)^{k+o(k)}.$$

The proof combines a certified off-diagonal Ramsey rate with the retained-spine and parameterized-book interfaces of Yang and Mao. Its new analytic component is an exact-diagonal correlation certificate. We prove a half-line ratio inequality for a cubic root filter, reduce all two-dimensional sign patterns to one exact diagonal expression, enclose the compact interval by 512-bit ball arithmetic, and control the remaining half-line analytically. Exact rational arithmetic then yields $\mathcal{G}_2^{(3)}(330867/500000, 12366348252219/1250000000000)$. A retained-spine max–min transfer and a second interval certificate give an exponent gain of $3.4754 \cdot 10^{-6}$. The certified unrounded base is below 3.780685288379640114. Four separately structured numerical replays, with a common Arb containment dependency, cover the continuous domains. The result is asymptotic and does not provide an effective finite-k threshold or a global parameter optimum.

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1. Introduction

For positive integers k, ℓ, let $R(k, \ell)$ be the least n such that every red–blue colouring of K_n contains either a red K_k or a blue K_ℓ. The classical argument of Erdős and Szekeres gives $R(k, k) \leq 4^{k+o(k)}$ [ES35]. Campos, Griffiths, Morris and Sahasrabudhe obtained the first exponential improvement of this base [CGMS23]. Gupta, Ndiaye, Norin and Wei subsequently reorganized the book method and stated the concrete bound $R(k, k) \leq 3.8^{k+o(k)}$ [GNNW24]. Yang and Mao introduced a retained-spine refinement in a multicolour setting [YM26].

This paper combines two ingredients. The first is a certified, symmetric off-diagonal rate

$$\log R(k, \ell) \leq kU(\ell/k) + o(k), \qquad 1 \leq \ell \leq k, \tag{1.1}$$

where $U = F_6$ is given explicitly in Section 3. The second is a new correlation certificate designed for the retained-spine transfer. The certificate is analytic except on compact intervals, where every inequality is proved by outward-rounded Arb balls [Joh17].

1.1. **Main result.** We state the result with its imported interfaces visible. The exact interfaces are recorded in Section 2; they are theorem statements from Yang–Mao together with the reviewed rate theorem (1.1), rather than unstructured assumptions about an implementation.

Theorem 1.1 (Conditional main theorem)**.** *Assume the version-pinned regularization, parameterized-book, and spine-compatibility interfaces stated in Imported interface 2.2, and the certified rate theorem Theorem 3.1. The positive-moment and tail implication used to establish the correlation input are proved locally in Section 4. Then*

$$R(k,k) \le \exp\Big((U(1) - 3.4754 \cdot 10^{-6})k + o(k)\Big) \le (3.780685290)^{k+o(k)}.$$

The certified unrounded upper base is less than 3.780685288379640114*.*

The paper is a source-relative proof package: every imported statement is named, every new inequality is proved here, and every numerical step is replayable by separately structured implementations. Those implementations share Arb's interval-containment semantics as a trust dependency. No finite value of the asymptotic threshold is claimed.

1.2. **The exact-diagonal mechanism.** Fix

$$u_0 = \frac{1235783}{500000}, \qquad L = u_0^3, \qquad a = e^{-u_0},$$

and define

$$E(z) = \sum_{n\ge 0} \frac{z^n}{(3n)!}, \qquad H(z) = 1 + aE(z)^2, \qquad G(z) = \frac{H(z) - H(-z)}{2}.$$

The correlation form is

$$F(x,y) = G(x)H(y) + G(y)H(x).$$

Our central observation is that the complete two-dimensional problem reduces to the exact identity

$$F(z,z) = H(z)^2 - H(z)H(-z), \qquad z \ge 0,$$

not merely to the larger square difference $H(z)^2 - H(-z)^2$. This improves the compact envelope while preserving a simple analytic tail.

1.3. **Proof architecture.** The proof has four logically separate layers.

(i) Section 3 states the off-diagonal rate and records its exact polynomial and verification boundary.
(ii) Section 4 proves the ratio/separator, the two-dimensional reduction, the exact diagonal envelope and the moment/tail budget.
(iii) Section 5 proves the retained-spine max–min transfer and inserts the certified parameters.
(iv) Section 6 describes continuum coverage, independent implementations and the immutable evidence manifest.

The complete repaired book induction and the frozen rate record are included in the appendices. Search scripts are not part of the proof; they only chose the exact rational data subsequently verified from scratch.

2. Ramsey rates and imported interfaces

2.1. **Symmetric homogeneous rate.** Let $U : [0,1] \to \mathbb{R}$ be continuous with $U(0) = 0$. Define its symmetric homogeneous extension by

$$\widehat{U}(x,y) = \begin{cases} \max\{x,y\}\, U\!\left(\dfrac{\min\{x,y\}}{\max\{x,y\}}\right), & \max\{x,y\} > 0, \\ 0, & x = y = 0. \end{cases} \tag{2.1}$$

For residual problems only, we extend the notation by $R(a,b) = 1$ when $\min\{a,b\} \le 0$; this means that no further page vertices are needed in at least one colour. It is not a new convention for positive Ramsey numbers.

Lemma 2.1 (Uniform homogeneous extension)**.** *Assume in addition that $U \ge 0$. Suppose (1.1) holds uniformly for $1 \le \ell \le k$, and the same bound is available after exchanging the two colours. Then, uniformly for $x, y \in [0,1]$,*

$$\log R(\lfloor xk \rfloor, \lfloor yk \rfloor) \le k\widehat{U}(x,y) + o(k). \tag{2.2}$$

Proof. The extension $\widehat{U}$ is continuous on the square (continuity at the origin uses $U(0) = 0$), hence uniformly continuous. Fix $\epsilon > 0$, and choose K_0 from the uniform epsilon formulation of (1.1). For $x \ge y$, put $k' = \lfloor xk \rfloor$ and $\ell' = \lfloor yk \rfloor$. If $k' \ge K_0$ and $\ell' \ge 1$, then

$$\log R(k', \ell') \le k'U(\ell'/k') + \epsilon k' = k\widehat{U}(k'/k, \ell'/k) + \epsilon k.$$

Uniform continuity and the $O(k^{-1})$ perturbation of the normalized coordinates replace the last $\widehat{U}$ by $\widehat{U}(x,y) + o(1)$, uniformly. If $k' < K_0$, only finitely many positive Ramsey numbers occur, so their logarithms are $O_{K_0}(1) = o(k)$; nonnegativity of $\widehat{U}$ permits the same upper bound. If one floored coordinate is zero, the residual convention gives logarithmic cost zero. The case $y \ge x$ follows by symmetry. Since ϵ is arbitrary, these cases give one uniform $o(k)$. ☐

2.2. **Yang–Mao interfaces.** The following package is extracted from the v1 source of Yang–Mao [YM26]. Its source member and line map are frozen in Appendix A. We state the complete specialization used later, so that the conditional theorem is falsifiable without guessing what the words "book interface" mean.

For completeness, we fix the precise meaning of the correlation input. We write $\mathcal{G}_2^{(3)}(\beta, C)$ when the following assertion holds: for every finite set $\mathcal{X}$, independent identically distributed $U, U' \in \mathcal{X}$, real Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, and maps $\sigma_i : \mathcal{X} \to \mathcal{H}_i$, if $Z_i = \langle \sigma_i(U), \sigma_i(U') \rangle$, then there are $i \in \{1,2\}$ and $\lambda \ge -1$ such that

$$\mathbb{P}\big(Z_i \ge \lambda,\ Z_{3-i} \ge -1\big) \ge \beta \exp\!\Big(-C(\lambda+1)^{1/3}\Big). \tag{2.2a}$$

For $0 < \beta \le 1$, $C > 0$, and parameters $p, \delta, \lambda_0 > 0$, put

$$L_\delta = \log(1/\delta), \qquad L_p = \log(1/p), \qquad \rho_\beta = \log(2/\beta),$$

$$\Pi = \frac{3\delta}{p} + \frac{6L_\delta L_p}{\lambda_0}, \qquad q = L_p + \Pi, \tag{2.3}$$

and

$$\Xi = 2\rho_\beta + 4C\lambda_0^{1/3} + \frac{12CL_\delta}{\lambda_0^{2/3}}. \tag{2.4}$$

Imported interface 2.2 (Retained-spine source interface)**.** Fix $0 < \eta$, $0 < p < 1/2 - \eta$, $0 < \delta \le \min\{p/4, 1/4\}$, $\lambda_0 \ge \max\{2, 6\log(1/\delta)\}$, and $\tau > 0$. Suppose the correlation property $\mathcal{G}_2^{(3)}(\beta, C)$ holds. The version-pinned Yang–Mao specialization supplies the following conclusions for all sufficiently large k.

(a) For every positive integer N, every red–blue colouring of K_N admits pairwise disjoint sets S_R, S_B, W such that S_R is a red clique, S_B is a blue clique, every edge from S_R to W is red, and every edge from S_B to W is blue. Writing $s_i = |S_i|$, one has

$$|W| \ge \left(\frac{1+\eta}{2}\right)^{s_R+s_B} N \tag{2.5}$$

and every $w \in W$ has at least $(1/2 - \eta)|W| - 1$ red neighbours and at least that many blue neighbours inside W.

(b) Let t, m be positive integers with $t \ge \lambda_0 \delta^{-1/2}$. If every $w \in W$ has at least $p|W|$ neighbours of each colour inside W, and

$$|W| \ge p^{-t} e^{\Pi t} m, \qquad |W| \ge 4t\, e^{2t\Xi}, \tag{2.6}$$

then the parameterized-book theorem, applied with $X = Y_R = Y_B = W$, returns a colour $i \in \{R, B\}$, a monochromatic t-clique $T \subseteq W$ of colour i, and $P_0 \subseteq W \setminus T$, $|P_0| \ge m$, such that every edge from T to P_0 has colour i.

(c) The compatibility conclusion is the following exact residual statement. If $i = R$, then a red K_{k-s_R-t} in P_0 (when this order is positive) extends with $S_R \cup T$, while a blue K_{k-s_B} extends with S_B. If $i = B$, interchange the colours, so the two residual orders are $k - s_R$ and $k - s_B - t$. A nonpositive residual order means that the retained spines already contain the required K_k.

For the proof below the exact correlation input is

$$\beta = \frac{330867}{500000}, \qquad C = \frac{12366348252219}{1250000000000}. \tag{2.7}$$

The point of stating Imported interface 2.2 is auditability. The correlation property in (2.7) is proved in this paper; the regularization and book compatibility are imported. The source line mapping and pinned archive hash are included in the reproducibility record.

2.3. **A general retained-spine transfer.** Put

$$c_\eta = \log \frac{2}{1+\eta}.$$

For a normalized spine vector $\sigma = (\sigma_R, \sigma_B) \in [0,1]^2$, define

$$D(\sigma) = c_\eta(\sigma_R + \sigma_B) + \widehat{U}(1-\sigma_R, 1-\sigma_B). \tag{2.8}$$

Here q and Ξ are the explicit quantities in (2.3)–(2.4). Define

$$\begin{aligned} M_\tau(\sigma) = \max\{&\widehat{U}((1-\sigma_R-\tau)_+, 1-\sigma_B), \\ &\widehat{U}(1-\sigma_R, (1-\sigma_B-\tau)_+)\}, \end{aligned} \tag{2.9}$$

$$P(\sigma) = c_\eta(\sigma_R+\sigma_B) + \tau q + M_\tau(\sigma), \tag{2.10}$$

$$Q(\sigma) = c_\eta(\sigma_R+\sigma_B) + 2\tau\Xi, \tag{2.11}$$

$$B(\sigma) = \max\{P(\sigma), Q(\sigma)\}. \tag{2.12}$$

Finally set

$$C_* = \max_{\sigma\in[0,1]^2} \min\{D(\sigma), B(\sigma)\}. \tag{2.13}$$

Theorem 2.3 (Retained-spine transfer)**.** *Under Imported interface 2.2 and the uniform rate (2.2),*

$$R(k,k) \le \exp(C_* k + o(k)).$$

Proof. Let $N = \lceil e^{(C_*+\varepsilon)k}\rceil$ and apply regularization. If either preliminary spine already has order at least k, there is nothing to prove. Otherwise write $s_i = \sigma_i k + O(1)$. From (2.5),

$$\log|W| \ge (C_*+\varepsilon)k - c_\eta(s_R+s_B) + o(k). \tag{2.14}$$

At this σ, choose the smaller of the direct and book branches.

If $D(\sigma) \le B(\sigma)$, then $D(\sigma) \le C_*$. Equations (2.2), (2.8) and (2.14) imply $|W| \ge R(k-s_R, k-s_B)$. The resulting red or blue clique extends with the corresponding preliminary spine.

If $B(\sigma) < D(\sigma)$, then both $P(\sigma) \le C_*$ and $Q(\sigma) \le C_*$. The former supplies the page size needed for either possible book colour, using (2.2) and (2.9); the latter supplies the common reservoir threshold. More explicitly, with $t = \lfloor \tau k \rfloor$ and m the larger of the two residual Ramsey thresholds, the two strict exponent inequalities give exactly (2.6). Since $p < 1/2 - \eta$ and the reservoir is exponential, the additive -1 in the degree conclusion is absorbed; fixed $\tau > 0$ also gives $t \ge \lambda_0 \delta^{-1/2}$. The parameterized-book interface produces the t-spine and a page set of size at least m, and compatibility extends the appropriate page clique with the retained spines to a monochromatic K_k. The fixed ε-slack absorbs every uniform $o(k)$ term. □

3. The certified off-diagonal rate

Set

$$h(\lambda) = (1+\lambda)\log(1+\lambda) - \lambda\log\lambda$$

and

$$\begin{aligned}
P_6(\lambda) = {} & -0.250000000000000\lambda + 0.019891840059418\lambda^2 \\
& - 0.012275954247190\lambda^3 + 0.144110816398083\lambda^4 \\
& + 0.006277913420654\lambda^5 - 0.066101623491668\lambda^6 \\
& - 0.002287675993602\lambda^7 - 0.058238059505066\lambda^8 \\
& + 0.030675864198693\lambda^9 + 0.052472201910597\lambda^{10} \\
& + 0.043300454861853\lambda^{11} - 0.042529975074653\lambda^{12} \\
& - 0.055263639426635\lambda^{13} + 0.036695886931268\lambda^{14}.
\end{aligned} \tag{3.1}$$

All displayed decimals in (3.1) are exact. Define

$$U(\lambda) = F_6(\lambda) = h(\lambda) + P_6(\lambda)e^{-\lambda}. \tag{3.2}$$

Theorem 3.1 (Certified rate, conditional on the stated trust boundary)**.** *For $1 \le \ell \le k$, uniformly in the ratio $\lambda = \ell/k$,*

$$\log R(k,\ell) \le kU(\lambda) + o(k), \tag{3.3}$$

and the same statement holds after exchanging k and ℓ. Moreover

$$e^{U(1)} = 3.78069842779612369875188164975221 62\ldots. \tag{3.4}$$

Proof. The certificate schema, all continuum obligations, and the theorem-to-rate implication are stated and proved in Proposition C.2. The exact-linked replay accepts all six certificate files, proves the elementary first prior, and verifies exact target-to-prior equality at the other five handoffs. Its sixth report gives $F_6(1) = 1.32990876182199307231878126 32\ldots$; outward-rounded exponentiation gives (3.4). The only nonlocal trust assumptions in this argument are the repaired BookCor induction reproduced in Appendix B and Arb's containment semantics. □

The proof is an exact-linked six-stage descent built on the repaired book-induction lemma. At each stage an interval certificate verifies the prior concavity, both Ramsey-region orientations, the main exponent inequality and the complete small-ratio interval. The first three stages use 65,536 cells; the last three use 131,072 cells. The sixth-stage minimum main slack is $4.81847 \cdot 10^{-5}$, while an independently implemented direct exponent check gives a minimum standard-orientation margin $1.06281 \cdot 10^{-6}$.

For the reader who wishes to separate mathematics from computation, the logical content is as follows.

(i) The combinatorial book induction converts a strict interior point of the asymptotic Ramsey region into a one-step rate improvement.

(ii) The analytic descent lemma is degree-independent: it requires only a concave prior rate and strict inequalities on the continuum.
(iii) Every prior polynomial equals the preceding target polynomial as an exact decimal tuple; no refitting occurs between stages.
(iv) Arb intervals enclose the continuum, not a sampled mesh. Search meshes are used only to propose a target.

The complete standalone book proof, certificate-correctness theorem, and stage-six record appear in Appendices B and C. The six JSON certificates and their hashes are part of the machine-readable evidence package.

4. The exact-diagonal correlation certificate

We now prove the new analytic input. Recall

$$u_0 = \frac{1235783}{500000}, \quad L = u_0^3, \quad a = e^{-u_0},$$

and

$$E(z) = \sum_{n\geq 0} \frac{z^n}{(3n)!}, \quad H(z) = 1 + aE(z)^2, \quad G(z) = \frac{H(z) - H(-z)}{2},$$

$$F(x,y) = G(x)H(y) + G(y)H(x).$$

4.1. **The bad-event separator.** For $u \geq 0$, define the cubic filters

$$P(u) = \frac{e^u + 2e^{-u/2}\cos(\sqrt{3}u/2)}{3}, \tag{4.1}$$

$$N(u) = \frac{e^{-u} + 2e^{u/2}\cos(\sqrt{3}u/2)}{3}. \tag{4.2}$$

Then $E(u^3) = P(u)$ and $E(-u^3) = N(u)$. Put

$$\sigma_0 = 10^{-8}, \qquad T = 2 + 2\sigma_0, qquad\sigma_* = \frac{1+2\sigma_0}{2} = \frac{50000001}{100000000}. \tag{4.3}$$

Lemma 4.1 (Half-line ratio)**.** *For every $u \geq u_0$,*

$$\frac{1 + aP(u)^2}{1 + aN(u)^2} > T. \tag{4.4}$$

Proof. Let $R(u) = (1 + aP(u)^2)/(1 + aN(u)^2)$. Its derivative has the sign of

$$W(u) = PP'(1 + aN^2) - NN'(1 + aP^2). \tag{4.5}$$

An outward-rounded enclosure proves $R(u_0) - T > 2.6595 \cdot 10^{-7}$ and $W > 0$ on 65,536 rational cells covering $[u_0, 2.9]$.

For $u \geq 2.9$, use

$$P(u) \geq \frac{e^u - 2e^{-u/2}}{3}, qquad|N(u)| \leq \frac{2e^{u/2} + e^{-u}}{3}.$$

After clearing the positive denominator, it suffices to prove positivity of

$$B_T(u) = e^{2u} - 4Te^u - 4e^{u/2} - 4Te^{-u/2} + 4e^{-u} - Te^{-2u}.$$

At $u = 2.9$, interval arithmetic proves $aB_T - 9(T-1) > 0$. Moreover

$$B_T'(u) \geq D(u) := 2e^u(e^u - 2T) - 2e^{u/2} - 4e^{-u},$$

and

$$D'(u) = 4e^{2u} - 4Te^u - e^{u/2} + 4e^{-u} > 0$$

throughout the half-line by exponential dominance, with a strict endpoint check. Hence B_T, and therefore $R - T$, remain positive. ☐

Lemma 4.2 (Separator)**.** *If one of A_1, A_2 is less than -1, then*

$$F(LA_1, LA_2) < -\sigma_*. \tag{4.6}$$

Proof. Let $x = -t$ with $t \geq L$, put $A = H(-t)$, and set $R = H(t)/H(-t)$. The exact identity

$$F(-t, y) = \frac{A}{2}\big((2-R)H(y) - H(-y)\big) \tag{4.7}$$

holds for every real y. By Lemma 4.1, $R > 2+2\sigma_0$, whereas $A, H(y), H(-y) \geq 1$. Thus the right side of (4.7) is less than $-(1+2\sigma_0)/2 = -\sigma_*$. Exchange the coordinates if necessary. ☐

4.2. **Complete two-dimensional reduction.**

Lemma 4.3 (Diagonal reduction)**.** *If $x, y \geq -L$ and $m = \max\{x, y, 0\}$, then*

$$F(x, y) \leq F(m, m). \tag{4.8}$$

Proof. The series for H has nonnegative coefficients, so H, and therefore its odd part G, are nondecreasing on the positive half-line. If $x = -v^3 \in [-L, 0]$, the root-filter formula gives

$$|E(-v^3)| \leq e^{v/2}, \qquad H(x) \leq 1 + e^{-u_0}e^v \leq 2. \tag{4.9}$$

Assume $x \leq y$. If $y < 0$, both odd parts are negative and hence $F(x,y) < 0 = F(0,0)$. If $x < 0 \leq y$, discard the nonpositive summand and use (4.9):

$$F(x,y) \leq G(y)H(x) \leq 2G(y) \leq 2G(y)H(y) = F(y,y).$$

If $0 \leq x \leq y$, monotonicity bounds both summands by $G(y)H(y)$. This exhausts all sign patterns and their boundaries. ☐

On the diagonal we retain the exact identity

$$F(z,z) = 2G(z)H(z) = H(z)^2 - H(z)H(-z), \qquad z \geq 0. \tag{4.10}$$

4.3. **Compact interval and analytic tail.** Let $z = u^3$ and $w = (u^3+u_0^3)^{1/3}$. Define

$$D_0 = \frac{88053}{10^9}.$$

Lemma 4.4 (Exact diagonal envelope)**.** *For every $u \geq 0$,*

$$\frac{H(u^3)^2 - H(u^3)H(-u^3)}{e^{4w}} < D_0. \tag{4.11}$$

Proof. The compact interval $[0, 20]$ is covered by 131,072 exact rational cells. On each cell a 512-bit Arb evaluation encloses the left side of (4.11); the minimum slack is greater than $4.2687 \cdot 10^{-6}$.

For $u \geq 20$, use $H(-u^3) \geq 1$ and write $H(u^3) = 1 + aP(u)^2$. Then

$$H(u^3)^2 - H(u^3)H(-u^3) \leq aP(u)^2 + a^2P(u)^4.$$

Since

$$|P(u)| \leq \frac{e^u}{3}(1 + 2e^{-3u/2})$$

and $w \geq u$, the normalized expression is at most

$$\frac{a}{9}e^{-2u}(1 + 2e^{-3u/2})^2 + \frac{a^2}{81}(1 + 2e^{-3u/2})^4. \tag{4.12}$$

Both terms decrease. At $u = 20$, (4.12) is below $8.805216326983 \cdot 10^{-5}$, leaving strict slack greater than $8.3673 \cdot 10^{-10}$; its limiting value $a^2/81$ has the same strict ordering. This proves the entire half-line. □

Combining Lemmas 4.3 and 4.4, if $A_1, A_2 \geq -1$ and $M = \max(A_1, A_2)$, then

$$F(LA_1, LA_2) < D_0 \exp\Big(4u_0(M + 1)^{1/3}\Big). \tag{4.13}$$

4.4. **Exact moment/tail budget.** The final passage from the separator to a correlation tail is short but proof-critical, so we give it in full rather than treating it as an unnamed interface.

Lemma 4.5 (Positive-moment expectation)**.** *For any finite set $\mathcal{X}$, independent identically distributed $U, U' \in \mathcal{X}$, real Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, and maps $\sigma_i : \mathcal{X} \to \mathcal{H}_i$, put $Z_i = \langle \sigma_i(U), \sigma_i(U') \rangle$. Then*

$$\mathbb{E}F(LZ_1, LZ_2) \geq 0. \tag{4.14}$$

Proof. The Taylor coefficients of E and H are nonnegative, and taking the odd part of H merely retains its odd-degree coefficients. Hence every bivariate Taylor coefficient of $F(x, y) = G(x)H(y) + G(y)H(x)$ is nonnegative. For nonnegative integers r, s, tensorization gives

$$\mathbb{E}(Z_1^r Z_2^s) = \big\|\mathbb{E}(\sigma_1(U)^{\otimes r} \otimes \sigma_2(U)^{\otimes s})\big\|^2 \geq 0.$$

Because $\mathcal{X}$ is finite, (Z_1, Z_2) has bounded support; the entire Taylor series is therefore absolutely integrable and may be averaged term by term. Summing the nonnegative terms proves (4.14). □

Take

$$\beta = \frac{330867}{500000}, \qquad \varepsilon = \frac{6893}{10^7}.$$

Exact rational arithmetic gives

$$\sigma_*(1 - \beta) - D_0\beta(1 + 2/\varepsilon) = \frac{39437634699447}{3446500000000000000} > 10^{-5}. \tag{4.15}$$

Proposition 4.6 (Certified correlation tail)**.** *The exact separator, envelope, and budget above imply*

$$\mathcal{G}_2^{(3)}\left(\frac{330867}{500000}, 4u_0(1+\varepsilon)\right).$$

Proof. Fix data from the definition (2.2a), put $\mathcal{E} = \{Z_1 \geq -1, Z_2 \geq -1\}$, $M = \max\{Z_1, Z_2\}$, $Y = ((M+1)_+)^{1/3}$, and $c = 4u_0$. By Lemmas 4.2 and 4.5 and (4.14), while (4.13) controls the good event,

$$0 \leq \mathbb{E}F(LZ_1, LZ_2) < D_0\mathbb{E}(e^{cY}\mathbf{1}_{\mathcal{E}}) - \sigma_*\mathbb{P}(\mathcal{E}^{\mathrm{c}}). \tag{4.16}$$

If $\mathbb{P}(\mathcal{E}) \geq \beta$, choose $\lambda = -1$ and either coordinate in (2.2a). Otherwise suppose, for a contradiction, that the claimed correlation conclusion fails for both coordinates at every $\lambda \geq -1$. For $v \geq 0$, taking $\lambda = v^3 - 1$ and using the union bound gives

$$\mathbb{P}(\mathcal{E} \cap \{Y \geq v\}) < 2\beta e^{-c(1+\varepsilon)v}. \tag{4.17}$$

The layer-cake identity and Tonelli's theorem now give

$$\begin{aligned}\mathbb{E}(e^{cY}\mathbf{1}_{\mathcal{E}}) &= \mathbb{P}(\mathcal{E}) + c\int_0^\infty e^{cv}\mathbb{P}(\mathcal{E} \cap \{Y > v\})\,dv \\ &< \beta + 2\beta c\int_0^\infty e^{-c\varepsilon v}\,dv = \beta(1 + 2/\varepsilon).\end{aligned}$$

Substitution into (4.16) makes its right-hand side at most $D_0\beta(1+2/\varepsilon) - \sigma_*(1-\beta) < 0$, by (4.15). This contradiction proves the proposition. □

Finally,

$$4u_0(1+\varepsilon) = \frac{12366348252219}{1250000000000} = 9.8930786017752,$$

which is the exact handoff asserted in (2.7).

5. The retained-spine certificate

The exact outer tuple is

$$\begin{aligned} &\eta = 0.02868896, \qquad p = 0.47130887, \qquad \delta = 0.000053863, \\ &\lambda_0 = 13233, \qquad \tau = 0.00069386, \qquad \Delta = 0.0000034754. \end{aligned} \tag{5.1}$$

Every terminating decimal in (5.1) is interpreted as a rational number.

The functions D, B and C_* are those of Theorem 2.3. Direct interval evaluation of the full square would introduce unacceptable variable dependence. The certificate instead uses the geometry of the ordered wedge. Concavity of U fixes the monotonicity of each page branch; the red and blue page expressions reduce to one-dimensional boundary intervals, and the diagonal reservoir reduces to an endpoint inequality. Every reduction is analytic before numerical evaluation.

Proposition 5.1 (Outer certificate)**.** *For the exact tuple (5.1) and the correlation input proved in Section 4,*

$$C_* \le U(1) - \Delta. \tag{5.2}$$

The certified lower margins are as follows.

Inequality	*lower margin*
degree gate	$1.5056 \cdot 10^{-9}$
red page	$5.1006 \cdot 10^{-9}$
blue page	$3.4825 \cdot 10^{-6}$
diagonal reservoir	$6.0468 \cdot 10^{-6}$

Proof. Write $x = \sigma_R$, $y = \sigma_B$, and by symmetry first assume $0 \le x \le y \le 1$. Put

$$A = U(1) - 2c_\eta, \qquad E = U'(1) - c_\eta.$$

The certificate proves $0 < A < E$. At $x = y = 1$, one has $U(1) - D(1,1) = A > \Delta$ because $r_d = \Delta/A < 1$, so the direct branch already applies. At every other point of the ordered triangle, $a = 1 - x \ge b = 1 - y$ has $a > 0$, and concavity at one gives

$$aU(b/a) \le aU(1) - (a-b)U'(1),$$

and hence

$$U(1) - D(x,y) \ge xA + (y-x)E. \tag{5.3}$$

Thus the direct branch applies outside the closed triangle

$$\mathcal{W} = \{0 \le x \le y : xA + (y-x)E \le \Delta\}.$$

Set

$$r_a = \Delta/E, \qquad r_d = \Delta/A, \qquad s = 1 - A/E.$$

The certified order $0 < r_a < \tau < r_d < 1$, $r_d + \tau < 1$, and $0 < s < 1$ shows that $\mathcal{W}$ has vertices $(0,0), (0,r_a), (r_d,r_d)$, with sloping side $y = r_a + sx$.

It remains to bound both possible page colours and the reservoir on $\mathcal{W}$. For the first page orientation put

$$z = \frac{1-\tau-x}{1-y}, \qquad P_R = c_\eta(x+y) + \tau q + (1-y)U(z).$$

On the relevant piece its coordinate derivatives are

$$\partial_x P_R = c_\eta - U'(z) < 0, qquad \partial_y P_R = c_\eta - U(z) + zU'(z) > 0.$$

It therefore suffices to follow the sloping side. Its derivative is

$$\mathcal{B}(z) = c_\eta - U'(z) + s\{c_\eta - U(z) + zU'(z)\},$$

where $\mathcal{B}'(z) = (sz-1)U''(z) > 0$. Along the side, z decreases from $z_a = (1-\tau)/(1-r_a)$; the 512-bit enclosure gives $\mathcal{B}(z_a) < -3.6952685 \cdot 10^{-4}$. Hence P_R is maximized at $(0, r_a)$, where

$$P_R(0,r_a) = c_\eta r_a + \tau q + (1-r_a)U\left(\frac{1-\tau}{1-r_a}\right). \tag{5.4}$$

For the other page orientation, the negative y-derivative reduces the maximum to $y = x$. There it equals

$$P_B(x,x) = 2c_\eta x + \tau q + (1-x)U\left(1 - \frac{\tau}{1-x}\right).$$

With $v = \tau/(1-x)$, its derivative is

$$\Gamma(v) = 2c_\eta - U(1-v) - vU'(1-v), \qquad \Gamma'(v) = vU''(1-v) < 0.$$

Thus the maximum occurs at $x = 0$; the enclosure gives $\Gamma(\tau) < -1.8476200 \cdot 10^{-4}$ and

$$P_B(0,0) = \tau q + U(1-\tau). \tag{5.5}$$

Finally, $Q(x,y) = c_\eta(x+y)+2\tau\Xi$ is coordinatewise increasing, so its maximum on $\mathcal{W}$ is

$$Q(r_d, r_d) = 2c_\eta r_d + 2\tau\Xi. \tag{5.6}$$

Subtracting (5.4)–(5.6) from $U(1) - \Delta$ gives, respectively, the red-page, blue-page, and reservoir margins in the table. The symmetric triangle $y \le x$ exchanges the two colours and adds no new inequality. The same run verifies the scalar source gates, including $\tau(1/2 - \eta - p) > 1.5056 \cdot 10^{-9}$, and the exact equality of the inner and outer correlation constants. Exact parameters, interval outputs, and checker identities are recorded in Appendix A. □

Proof of Theorem 1.1. By Theorem 2.3 and Proposition 5.1,

$$R(k,k) \le \exp((U(1) - \Delta)k + o(k)).$$

A 512-bit Arb evaluation gives

$$\begin{aligned} e^{U(1)-\Delta} &< 3.7806852883796401133524 \\ &< 3.780685288379640114 < 3.780685290, \end{aligned}$$

where the first decimal is an outward-rounded upper endpoint. Its distance to the safe nine-decimal value is greater than $1.6203 \cdot 10^{-9}$. This proves both printed claims. □

6. Certified computation and independent replay

The computational proof follows three rules.

Exact data. All polynomial coefficients and parameters are parsed from decimal or rational strings, never from binary floating-point literals. Equality of the inner and outer correlation constants is checked in exact rational arithmetic.

Continuum coverage. Every compact domain is a union of closed rational intervals. Arb evaluates an inclusion extension on each cell. Analytic monotonicity or derivative signs treat the unbounded tails. A dense floating scan is not proof and is used only during search.

Independent implementations. The evidence package contains two author checkers, a non-importing replay, and an adaptive referee replay. The latter two reconstruct the functions and constants instead of importing author code. They use different subdivisions and endpoint reductions. All four use 512-bit precision and `python-flint==0.9.0`.

The current theorem files have the following SHA-256 identifiers.

Role	SHA-256 prefix
proof specification	`e4ea58def6405936`
inner checker	`17db00d9374ce3ba`
outer checker	`e2035cbffefcb147`
non-importing replay	`fc379e3b861b6905`
adaptive replay	`3c30f72ddc24848f`
independent proof review	`a21402a5dad40490`

From the repository root, a lightweight replay of the exact-diagonal and outer-transfer certificates is

```
bash scripts/bootstrap.sh
bash scripts/reproduce_upper.sh
```

The first command creates a Python 3.11 environment with exact dependency versions. The second runs the four final transfer checkers; it deliberately does not replay the six-stage rate chain. The complete upper-bound replay, including all six rate certificates, both Ramsey-region orientations, the combinatorial audit, and all four final transfer checkers, is

```
.venv/bin/python reproduce.py full
```

The unified manuscript is compiled separately with `make paper`.

7. Scope, trust boundary and further work

The theorem is asymptotic. The regularization and book interfaces contain eventual estimates, and this paper does not extract an effective threshold for k. Consequently the result has no direct implication for a specific finite Ramsey number.

The retained-spine parameters were found by bounded numerical search and then frozen. The proof certifies the frozen point; it does not prove global optimality. A better correlation inequality, a different off-diagonal rate, or a richer outer partition may improve the decimal further.

The external trust boundary consists of the explicitly stated Yang–Mao v1 interfaces, the repaired local book/rate theorem recorded in the appendices, and the containment semantics of Arb. Independent implementations reduce the risk of a transcription or dependency error but are not a proof-assistant formalization.

Finally, the numerical comparison in the introduction is made against published or source-stated bounds that are cited explicitly. This draft does not claim priority, novelty against all unpublished work, acceptance, or an unqualified world-best status.

Appendix A. Exact certificate data

The proof specification is the immutable file `routes/upper/EXACT_DIAGONAL_NEXT_CANDIDATE.md`. Its SHA-256 is

`e4ea58def640593690a7545e111c3b38f1bfcf8a5735fe7985600481e0bf36d4`.

The exact constants are

$$u_0 = 1235783/500000, \quad \sigma_0 = 10^{-8}, \quad D_0 = 88053/10^9,$$

$$\begin{aligned} \beta &= 330867/500000, & \varepsilon &= 6893/10^7, \\ C &= 12366348252219/1250000000000. \end{aligned}$$

The compact ratio proof uses 65,536 cells on $[u_0, 2.9]$; the exact diagonal uses 131,072 cells on $[0, 20]$. Both tails are proved by explicit monotone envelopes. The outer checker verifies the full ordered wedge and the final decimal rounding.

The repository integrity checker validates all six frozen proof files before running any expensive enclosure. In particular, a successful numerical run against a modified proof specification is not accepted as a reproduction of this theorem.

A.1. **Pinned source interface.** The imported source member is `main.tex` from Yang–Mao v1, arXiv:2608.01962v1. Its SHA-256 is

`155b7104ec5b6935a576ae9f2b161976a966b0b46bd2b69153c0934ca688da2a`.

The exact mapping for the conclusions actually imported in Imported interface 2.2 is:

source lines	imported conclusion
318–332	preliminary colour spines, common reservoir, reservoir-size lower bound, and minimum colour degrees
1146–1565	parameterized monochromatic-book theorem and proof with the exact ρ_β, Π, Ξ constants
1569–1618	compatibility of preliminary spines with the final book

For source lineage, but not as additional imported assumptions, the analytic antecedents are:

source lines	locally used lineage
370–418	positive tensor-moment lemma, reproved in the specialized form Lemma 4.5
426–791, 793–983	generic root-filter, separator, and expectation–tail argument; the stronger exact-diagonal specialization is proved in Section 4
989–1009	definition and parameter domain of $\mathcal{G}_2^{(3)}(\beta, C)$, restated in (2.2a)
1017–1142	density-increment lemma internal to the pinned source proof of the imported book theorem

The source archive is downloaded on demand and checked against this digest; it is not silently replaced by a later arXiv version. The manuscript proves the stronger local correlation input (2.7); only the regularization, parameterized-book, and compatibility conclusions in the first table are imported.

A.2. **Outer-wedge certificate data.** All outer parameters are exact terminating rationals. Substitution in (2.3)–(2.4) gives outward-rounded enclosures

$$q \in [0.7559369286692751849342481567047,\\ 0.7559369286692751849342481567048],$$

$$\Xi \in [940.3014597301789400852772355749,\\ 940.3014597301789400852772355750].$$

The proof-critical lower or upper endpoints are:

quantity	certified endpoint
$\sup_{(0,1]} U''$	< -0.3602795262399473
degree slack	$> 1.5056762000000000 \cdot 10^{-9}$
red boundary derivative	$< -3.6952685632128072 \cdot 10^{-4}$
red page margin	$> 5.1006707822905727 \cdot 10^{-9}$
blue diagonal derivative	$< -1.8476200284472833 \cdot 10^{-4}$
blue page margin	$> 3.4825658438202073 \cdot 10^{-6}$
reservoir margin	$> 6.0468779601096049 \cdot 10^{-6}$
base upper endpoint	$< 3.7806852883796401133524$
safe-decimal rounding margin	$> 1.6203598866476283 \cdot 10^{-9}$

The primary outer checker has SHA-256

`e2035cbffefcb147141fcee4831cac2af085f26d9183091ee20d90acc89ac87b`,

the non-importing replay has SHA-256

`fc379e3b861b69054aadaa80ebc3c791cb8358f22c9b0aa01070c56aa131c26c`,

and the adaptive referee replay has SHA-256

`3c30f72ddc24848f2b72c278f35b5d8bd6296ae8a40b045d30c5b7e087bd7150`.

They use different subdivision and endpoint-reduction structures, but all three trust python-flint/Arb containment and the same exact mathematical specification; they are not described as independent formal proofs.

Appendix B. Standalone proof of the GNNW book lemma

Date: 2026-08-12
Source being repaired: Gupta–Ndiaye–Norin–Wei, arXiv:2407.19026v1, Lemmas 7–11 and Theorem 12.

B.1. **Claim boundary.** This note proves the combinatorial input used by the certified diagonal Ramsey descent without invoking GNNW Lemma 11 or Theorem 12 as black boxes. It retains their definitions of a candidate and of the rate region, and it uses only elementary Ramsey recursion, convexity, averaging, and induction.

Let a **candidate** be a pair of nonempty disjoint vertex sets (X, Y) in a red–blue complete graph. Its red cross-density is

$$d(X, Y) = \frac{e_R(X, Y)}{|X||Y|}.$$

It is (k, ℓ, t)-good if $X \cup Y$ contains a red K_k, or X contains a blue K_t, or Y contains a blue K_ℓ.

Let $\mathcal{R}$ be the closure of the set of pairs $(x, y) \in (0, 1)^2$ for which, for all sufficiently large $k + \ell$,

$$R(k, \ell) \le x^{-k} y^{-\ell},$$

and let $\mathcal{R}_*$ be its interior.

B.1.1. *Theorem A (book induction).* Suppose $0 < \mu_0, x_0, y_0, p < 1$,

$$x_0 < p^{1/(1-\mu_0)}(1 - \mu_0), \qquad (x_0, y_0) \in \mathcal{R}_*. \tag{A.1}$$

There is L_0 such that, for all positive integers k, ℓ, t with $\ell \ge L_0$, every candidate of density at least p and satisfying

$$|X||Y| \ge x_0^{-k} y_0^{-\ell} \mu_0^{-t} \tag{A.2}$$

is (k, ℓ, t)-good.

B.1.2. *Corollary B (balanced-book bound).* Suppose $0 < \mu, x, y, p < 1$,

$$x < p^{1/(1-\mu)}(1 - \mu), \qquad (x, y) \in \mathcal{R}_*. \tag{B.1}$$

For all sufficiently large ℓ, every red–blue coloring of K_N of red density at least p, where

$$N \ge x^{-k/2} (\mu y)^{-\ell/2}, \tag{B.2}$$

contains a red K_k or a blue K_ℓ.

The statements hold for all $p, \mu \in (0, 1)$; no relation $p > \mu$ is needed.

B.2. **Four elementary inputs.**

B.2.1. *Lemma 1 (an interior point gives its own eventual bound).* If $(u, v) \in \mathcal{R}_*$, then, for all sufficiently large $a + b$,

$$R(a, b) \le u^{-a} v^{-b}. \tag{1.1}$$

Proof. Choose $\eta > 0$ so that $(u + 2\eta, v + 2\eta) \in \mathcal{R}$. Since $\mathcal{R}$ is the closure of the eventual-bound set, there is a point (u', v') in that set with $u' > u$ and $v' > v$. Its defining bound is stronger than (1.1). □

B.2.2. *Lemma 2 (density convexity).* For every candidate,

$$\sum_{v \in X} d(X, N_R(v) \cap Y) |N_R(v) \cap Y| \ge e_R(X, Y) d(X, Y), \tag{2.1}$$

where an empty-neighborhood summand is interpreted as zero.

Proof. Put $d_y = |N_R(y) \cap X|$. Multiplying the left side by $|X|$ gives $\sum_{y \in Y} d_y^2$. Cauchy–Schwarz bounds this below by $|Y|^{-1} (\sum_y d_y)^2 = e_R(X, Y)^2 / |Y|$, which is $|X|$ times the right side. □

B.2.3. *Lemma 3 (blue-book extraction).* Let $0 < \nu < 1$, and let b, k, m be positive integers with $m \ge 5\nu^{-1} b^2$ and $b \le m$. Suppose $|X| \ge 5m^2$ and at least $R(k, m)$ vertices $v \in X$ have blue degree at least $\nu |X|$ inside X. Then X contains a red K_k, or it contains disjoint sets S, T such that S is a blue clique, every edge from S to T is blue, and

$$|S| = b, \qquad |T| \ge \frac{\nu^b}{2} |X|. \tag{3.1}$$

Proof. The high-blue-degree vertices contain a red K_k or a blue K_m. In the latter case call its vertex set U. Put $d_z = |N_B(z) \cap U|$ for $z \in X \setminus U$ and let D be their average. Counting ordered blue pairs from U to $X \setminus U$ gives

$$D \ge \frac{m(\nu |X| - m)}{|X| - m} = \nu m - \frac{(1 - \nu) m^2}{|X| - m} > \nu m - \frac{1}{4}, \tag{3.2}$$

where $|X| \ge 5m^2$ and $m \ge 1$ were used in the last inequality. Let $q = \lfloor D \rfloor$. The discrete convexity of $j \mapsto \binom{j}{b}$ on nonnegative integers (with value zero for $j < b$) shows that a uniform b-subset S of U satisfies

$$\mathbb{E} |N_B(S) \cap (X \setminus U)| = \binom{m}{b}^{-1} \sum_{z \in X \setminus U} \binom{d_z}{b} \ge \binom{m}{b}^{-1} \binom{q}{b} |X \setminus U|. \tag{3.3}$$

Since $q > D - 1 > \nu m - 5/4$ and $m \ge 5\nu^{-1} b^2$, every $0 \le i < b$ satisfies

$$\frac{q - i}{m - i} > \nu - \frac{(1 - \nu) i + 5/4}{m - i} \ge \nu \left(1 - \frac{b + 1/4}{5b^2} \right).$$

For the second inequality, the fraction being subtracted increases with i, and at $i = b - 1$ it is at most the claimed value because

$$\frac{(1-\nu)(b-1)+5/4}{5b^2/\nu-(b-1)} \leq \frac{\nu(b+1/4)}{5b^2};$$

after cross-multiplication, the difference between the right numerator and the left numerator is $\nu(b-1)(1-(b+1/4)/(5b^2)) \geq 0$.

(If $q < b$, the last strict lower bound would be positive at $i = q$, an impossibility, so the binomial ratio is well-defined.) Hence

$$\frac{\binom{q}{b}}{\binom{m}{b}} \geq \nu^b \left(1 - \frac{b+1/4}{5b^2}\right)^b \geq \frac{3}{4}\nu^b. \tag{3.4}$$

The last elementary bound is weakest at $b = 1$, where its left factor is $3/4$; for $b \geq 2$, Bernoulli's inequality gives at least $1-(b+1/4)/(5b) > 3/4$. Finally $|X \setminus U| \geq (4/5)|X|$, so (3.3)–(3.4) exceed $3\nu^b|X|/5 > \nu^b|X|/2$. Some S therefore has a common blue page T satisfying (3.1). □

B.2.4. *Lemma 4 (the limit needed to choose the moment).* For every $p, \mu \in (0,1)$,

$$\lim_{r\to\infty} (p^{1/r} - \mu)^r (1-\mu)^{1-r} = p^{1/(1-\mu)}(1-\mu), \tag{4.1}$$

where r runs through sufficiently large positive integers, for which the base is positive.

Proof. Since $p^{1/r} \to 1 > \mu$, positivity is eventual, and

$$r \log \left(1 + \frac{p^{1/r}-1}{1-\mu}\right) \longrightarrow \frac{\log p}{1-\mu}.$$

Exponentiation and the remaining factor $1-\mu$ prove (4.1). □

B.3. **Proof of Theorem A.** The strict inequality in (A.1) implies $x_0 + \mu_0 < 1$. By Lemma 4 choose an integer $r > 1$ such that

$$x_0 < (p^{1/r} - \mu_0)^r (1-\mu_0)^{1-r}. \tag{5.1}$$

After fixing r, choose $\varepsilon > 0$ sufficiently small that

$$\begin{gathered} p \geq 2\varepsilon, \qquad (1+\varepsilon)(\mu_0+\varepsilon) \leq 1, \qquad x_0 + \mu_0 + 2\varepsilon < 1, \\ (x_0+2\varepsilon, y_0+2\varepsilon) \in \mathcal{R}_*, \\ \mu_0 + 2\varepsilon \leq \frac{\mu_0+\varepsilon}{(\mu_0+\varepsilon)+(x_0+\varepsilon)}, \qquad (p-\varepsilon)^{1/r} > \mu_0 + 3\varepsilon, \\ x_0 \leq \left((p-\varepsilon)^{1/r} - \mu_0 - 3\varepsilon\right)^r (1-\mu_0-\varepsilon)^{1-r} - \varepsilon. \end{gathered} \tag{5.2}$$

All requirements follow by openness, continuity, (5.1), and the strict inequality $x_0 + \mu_0 < 1$. Set

$$x = x_0 + \varepsilon, \qquad y = y_0 + \varepsilon, \qquad \mu = \mu_0 + \varepsilon, \qquad \delta_n = \frac{\varepsilon}{n}. \tag{5.3}$$

We prove the following stronger assertion by induction on $n = k + t$, with $\ell \geq L_0$ fixed. If

$$d(X, Y) \geq p - \delta_n \tag{5.4}$$

and

$$(d(X, Y) + \delta_n - p)^r |X||Y| \geq x^{-k} y^{-\ell} \mu^{-t}, \tag{5.5}$$

then (X, Y) is (k, ℓ, t)-good. The cases $k = 1$ or $t = 1$ are immediate.

First note that (A.2) and $d(X, Y) \geq p$ imply (5.5): because $x_0 \leq x/(1+\varepsilon)$, $y_0 \leq y/(1+\varepsilon)$, and $\mu_0 \leq \mu/(1+\varepsilon)$, it is enough that

$$\frac{\varepsilon^r}{n^r}(1+\varepsilon)^{n+\ell} \geq 1. \tag{5.6}$$

A single sufficiently large L_0 makes (5.6) true for every $n \geq 2$.

B.3.1. *Regularization and the size of X.* Delete from X any vertex having fewer than $(p - \delta_n)|Y|$ red neighbors in Y. Both the nonnegative excess

$$e_R(X, Y) - (p - \delta_n)|X||Y|$$

and its normalization by $|X||Y|$ do not decrease, so the left side of (5.5) does not decrease. The positive right side prevents deletion of all vertices. We may consequently assume

$$d(X', Y) \geq p - \delta_n \quad \text{for every nonempty } X' \subseteq X. \tag{5.7}$$

Lemma 1 and the interior condition in (5.2) imply, for sufficiently large L_0, that if

$$|Y| \geq (x + \varepsilon)^{-k}(y + \varepsilon)^{-\ell},$$

then Y already contains a red K_k or a blue K_ℓ. Otherwise, using $d + \delta_n - p \leq 1$ in (5.5),

$$|X| \geq \left(\frac{x+\varepsilon}{x}\right)^k \left(\frac{y+\varepsilon}{y}\right)^\ell \mu^{-t} \geq (1+\varepsilon)^{n+\ell}. \tag{5.8}$$

B.3.2. *The big-blue branch.* Put

$$b = \left\lceil \frac{2r\log n - r\log\varepsilon + \log 2}{\log(1+\varepsilon)} \right\rceil, \qquad m = \lceil 5\mu^{-1}b^2 \rceil, \qquad w = R(k, m). \tag{5.9}$$

Here $m = O_{r,\varepsilon}(\log^2 n)$, $m \geq b$, and $w = R(k, m) \leq \binom{k+m-2}{m-1} \leq (k+m)^m \leq \exp(O_{r,\varepsilon}(\log^3 n))$. Since (5.8) is exponential in $n + \ell$, increasing L_0 once makes, uniformly for every n,

$$|X| \geq 5m^2, \qquad w \leq \frac{\varepsilon(p-\varepsilon)}{n^3}|X|. \tag{5.10}$$

Let

$$W = \{v \in X : |N_B(v) \cap X| \geq (\mu + \varepsilon)|X|\}.$$

If $|W| \geq w$, Lemma 3, with $\nu = \mu + \varepsilon$, produces a red K_k or a blue book with base S of size b and page T of size at least $(\mu + \varepsilon)^b |X|/2$. If $b \geq t$, its base already contains a blue K_t. Suppose $b < t$. From (5.7), $d(T, Y) \geq p - \delta_n$, and hence

$$d(T, Y) + \delta_{n-b} - p \geq \delta_{n-1} - \delta_n \geq \frac{\varepsilon}{n^2}.$$

Using (5.5), $d + \delta_n - p \leq 1$, and the definition of b gives

$$\begin{aligned} &(d(T, Y) + \delta_{n-b} - p)^r |T||Y| \\ &\quad \geq \frac{1}{2} \left(\frac{\varepsilon}{n^2}\right)^r (\mu + \varepsilon)^b |X||Y| \\ &\quad \geq \frac{1}{2} \left(\frac{\varepsilon}{n^2}\right)^r \left(\frac{\mu + \varepsilon}{\mu}\right)^b x^{-k} y^{-\ell} \mu^{-t+b} \\ &\quad \geq x^{-k} y^{-\ell} \mu^{-t+b}. \end{aligned} \tag{5.11}$$

Induction applied to (T, Y) with parameter $t - b$ finishes this branch: a blue K_{t-b} in T joins the blue base S.

We may therefore assume

$$|W| < w. \tag{5.12}$$

B.3.3. *The red-step/density-increment branch.* Lemma 2 and (5.10)–(5.12) show that the contribution from W is at most

$$w|Y| \leq \frac{\varepsilon}{n^3} e_R(X, Y).$$

Consequently some $v \in X \setminus W$ satisfies

$$d(X, N_R(v) \cap Y) \geq d(X, Y) - \frac{\varepsilon}{n^3}. \tag{5.13}$$

Write

$$X_R = N_R(v) \cap X, \quad X_B = N_B(v) \cap X, \quad Y' = N_R(v) \cap Y, \quad p' = p - \delta_{n-1},$$

and

$$\alpha = d(X, Y') - p', \quad \alpha_R = d(X_R, Y') - p', \quad \alpha_B = d(X_B, Y') - p'.$$

An empty X_R or X_B is assigned zero weighted contribution and its induction branch is skipped; its corresponding α_R or α_B need not be defined. By (5.7), $|Y'| \geq (p - \varepsilon)|Y|$. Also, (5.4) and (5.13) give

$$\alpha \geq \delta_{n-1} - \delta_n - \frac{\varepsilon}{n^3} \geq \frac{\varepsilon}{n^2} > 0. \tag{5.14}$$

Since all edges from v to Y' are red,

$$\frac{\alpha_R}{\alpha}\frac{|X_R|}{|X|} + \frac{\alpha_B}{\alpha}\frac{|X_B|}{|X|} + \frac{1}{\alpha|X|} \geq 1. \tag{5.15}$$

If $\alpha_R \geq 0$ and

$$\alpha_R^r|X_R| \geq \frac{x\alpha^r|X|}{p-\varepsilon}, \tag{5.16}$$

then, using $|Y'| \geq (p-\varepsilon)|Y|$, (5.13), and

$$d(X,Y') - p' \geq d(X,Y) + \delta_n - p,$$

condition (5.16) and (5.5) show that (X_R, Y') satisfies the inductive moment inequality for $(k-1, \ell, t)$. Its goodness implies that of (X,Y) because v is red to $X_R \cup Y'$. The analogous condition, including its necessary sign hypothesis, is

$$\alpha_B \geq 0, \qquad \alpha_B^r|X_B| \geq \frac{\mu\alpha^r|X|}{p-\varepsilon} \tag{5.17}$$

gives the inductive branch $(k, \ell, t-1)$, since v is blue to X_B.

Assume both branches fail. Set

$$a = |X_R|/|X|, \qquad c = |X_B|/|X|, \qquad q = 1 - 1/r.$$

Using the strict failures of (5.16)–(5.17) in (5.15) gives

$$x^{1/r}a^q + \mu^{1/r}c^q + \frac{(p-\varepsilon)^{1/r}}{\alpha|X|} > (p-\varepsilon)^{1/r}. \tag{5.18}$$

Now $a + c = 1 - |X|^{-1} < 1$ and $c \leq \mu + \varepsilon$. The function

$$f(z) = x^{1/r}(1-z)^q + \mu^{1/r}z^q$$

is increasing on $0 \leq z \leq \mu/(\mu+x)$. The special smallness condition in (5.2) is exactly

$$\mu + \varepsilon \leq \frac{\mu}{\mu+x}.$$

Therefore

$$x^{1/r}a^q + \mu^{1/r}c^q \leq f(c) \leq f(\mu+\varepsilon) \leq x^{1/r}(1-\mu)^q + \mu + \varepsilon. \tag{5.19}$$

Equations (5.8) and (5.14), after one final uniform increase of L_0, give

$$\frac{(p-\varepsilon)^{1/r}}{\alpha|X|} \leq \frac{n^2}{\varepsilon(1+\varepsilon)^{n+\ell}} \leq \varepsilon. \tag{5.20}$$

Substituting (5.19)–(5.20) into the strict inequality (5.18) yields

$$x^{1/r}(1-\mu)^{1-1/r} + \mu + 2\varepsilon > (p-\varepsilon)^{1/r}.$$

The positivity condition in (5.2) allows the rearrangement

$$x > \left((p-\varepsilon)^{1/r} - \mu - 2\varepsilon\right)^r (1-\mu)^{1-r}. \tag{5.21}$$

But $x = x_0 + \varepsilon$ and $\mu = \mu_0 + \varepsilon$, so (5.21) strictly contradicts the final inequality of (5.2). The induction, and hence Theorem A, is complete. □

B.4. **Proof of Corollary B.** By openness choose $y_0 > y$ with $(x, y_0) \in \mathcal{R}_*$. Take a uniformly random equitable bipartition of the N vertices. Each edge has the same probability of crossing, so some partition (X, Y) has cross-red density at least the global red density p.

For all sufficiently large ℓ,

$$(y_0/y)^{\ell/2} \geq 3.$$

Thus (B.2) makes both equitable parts at least $x^{-k/2}(\mu y_0)^{-\ell/2}$, and hence

$$|X||Y| \geq x^{-k} y_0^{-\ell} \mu^{-\ell}.$$

Apply Theorem A with $t = \ell$. A red K_k in $X \cup Y$, a blue K_ℓ in X, or a blue K_ℓ in Y gives the desired conclusion. □

B.5. **Source defects neutralized by this proof.** The standalone proof explicitly repairs all of the following source-level issues rather than silently importing them:

1. Lemma 10 does not require $p > \mu$.
2. The moment order r is chosen before ε.
3. The northeast rate point is chosen in $\mathcal{R}_*$, and Lemma 1 supplies the exact eventual bound used later.
4. The printed comparison for y_0 uses x in place of y.
5. The case $b \geq t$ and empty X_R, X_B are treated separately.
6. The normalized sizes satisfy $a + c < 1$, not the dimensionally false printed inequality.
7. The concavity calculation supplies an upper bound in (5.19), and the strictness in (5.18) is preserved to obtain a genuine contradiction.

The only non-symbolic dependency left in the diagonal descent is the stated semantics of the Arb interval implementation used to certify its numerical inequalities. A post-repair adversarial replay of this standalone proof and a generated-copy fidelity audit have both passed. Independent external human review and public archival remain pending.

Appendix C. Sixth chained upper-bound descent: bounded search log

C.1. **Certificate semantics and correctness.** We first state the theorem-to-JSON implication used by all six links. For a finite exact-decimal coefficient tuple $c = (c_1, \dots, c_d)$, write

$$F_c(\lambda) = (1+\lambda)\log(1+\lambda) - \lambda\log\lambda + e^{-\lambda}\sum_{j=1}^{d} c_j\lambda^j. \tag{C.1}$$

A file with schema `corrected-two-sided-ramsey-v1` contains exact tuples $c^{\mathrm{prior}}, c^{\mathrm{target}}$, an exact splice λ_*, and a finite ordered list of records $(\lambda_i^-, \lambda_i^+, M_i, Y_i)$. Acceptance means that outward-rounded interval evaluation establishes all of the following obligations.

(i) The records form an exact partition of $[\lambda_*, 1]$, with $0 < M_i, Y_i < 1$, and the prior rate is strictly concave on $(0, 1]$.

(ii) On every record and every λ in its closed interval, put

$$X = (1 - M_i)\big(1 - e^{-F'_{c^{\mathrm{target}}}(\lambda)}\big)^{1/(1-M_i)}, \quad a = -\log X, \quad b = -\log Y_i.$$

The verifier proves $F_{c^{\mathrm{target}}} > 0$, $F'_{c^{\mathrm{target}}} > 0$, and both prior-region inequalities

$$a + \mu b > F_{c^{\mathrm{prior}}}(\mu), \qquad b + \mu a > F_{c^{\mathrm{prior}}}(\mu) \quad (0 < \mu \le 1). \tag{C.2}$$

Strict concavity reduces the two suprema in μ to their endpoint and unique stationary-point candidates; every candidate is enclosed.

(iii) On the same complete record interval the main descent slack satisfies

$$F_{c^{\mathrm{target}}}(\lambda) + \frac{1}{2}\{\log X + \lambda\log M_i + \lambda\log Y_i\} > 0. \tag{C.3}$$

(iv) On the open interval $(0, \lambda_*]$, the verifier takes $M = \lambda e^{-\lambda}$, defines X by the same formula and $Y = 1 - X$, and proves positivity of F, F' and a strictly positive lower bound for the left side of (C.3) divided by λ. Bounds $q_-\lambda \le e^{-F'} \le q_+\lambda$ cancel the two $\lambda\log\lambda$ singularities, so this is an analytic tail proof rather than extrapolation from λ_*.

(v) At the first link, the same elementary witness proves the prior rate. At every later link, c^{prior} agrees entry by entry, as exact decimal data, with the preceding accepted target tuple.

We next close the theorem-to-certificate bridge, including the two steps that are easiest to lose in a numerical presentation: the density dichotomy and the induction in the smaller clique order.

Lemma C.1 (Finite-net rate descent)**.** *Let $G : [0,1] \to \mathbb{R}$ be continuous, twice continuously differentiable on $(0,1]$, and satisfy $G(0) = 0$, $G, G' > 0$. Suppose $G(\lambda) = H(\lambda) + O(\lambda)$ at zero, where $H(\lambda) = (1+\lambda)\log(1+\lambda) - \lambda\log\lambda$. For every $\lambda \in (0,1]$, suppose there are $M, X, Y \in (0,1)$ with*

$$X = (1 - M)\big(1 - e^{-G'(\lambda)}\big)^{1/(1-M)} \tag{C.4}$$

and

$$G(\lambda) + \frac{1}{2}\{\log X + \lambda\log M + \lambda\log Y\} > 0. \tag{C.5}$$

Assume that on every compact $I \subset (0,1]$, $\inf_I G' > 0$, $\sup_I M < 1$, and, for every sufficiently small $\delta > 0$, the point

$$p_{\lambda,\delta} = 1 - e^{-G'(\lambda)+2\delta}, \qquad x_{\lambda,\delta} = e^{-\delta}(1-M)p_{\lambda,\delta}^{1/(1-M)}, \tag{C.6}$$

paired with Y, lies in $\mathcal{R}_*$. Then, uniformly for $1 \le \ell \le k$,

$$\log R(k,\ell) \le kG(\ell/k) + o(k). \tag{C.7}$$

Proof. Fix $\epsilon > 0$. The Erdős–Szekeres recursion and Stirling's formula give $\log R(k,\ell) \le kH(\ell/k) + o(k)$, uniformly as $\ell/k \to 0$. Since $H - G = O(\lambda)$, choose $\rho > 0$ so that the desired bound with error ϵk already holds when $\ell \le \rho k$. It remains to consider $I = [\rho, 1]$.

The hypotheses in the lemma make the logarithm of $x_{\lambda,\delta}/X(\lambda)$ converge uniformly to zero as $\delta \downarrow 0$. Choose one $\delta > 0$ such that, throughout I,

$$e^{-\epsilon}X(\lambda) \le x_{\lambda,\delta} < X(\lambda). \tag{C.8}$$

Uniform continuity of G, G' supplies a finite descending grid $\{\lambda_j\}$ such that every $r \in I$ has a $\lambda_j \in [r, r+h]$ with

$$|G(\lambda_j) - G(r)| \le \epsilon/2, qquad |G'(\lambda_j) - G'(r)| \le \delta. \tag{C.9}$$

Apply the balanced-book corollary of Appendix B at each grid point with $p = p_{\lambda_j,\delta}$, $\mu = M(\lambda_j)$, $x = x_{\lambda_j,\delta}$, and $y = Y(\lambda_j)$. The strict book hypothesis follows from (C.6), and the assumed interior condition supplies the other hypothesis. Since the grid is finite, the largest of the resulting thresholds is finite.

Now let $N \ge e^{k(G(r)+\epsilon)}$, where $r = \ell/k \in I$. If the red-edge density is at least $1 - e^{-G'(r)+\delta}$, it is at least $p_{\lambda_j,\delta}$ by (C.9). Equations (C.5), (C.8), (C.9), $\lambda_j k \ge \ell$, and $\log(MY) < 0$ give

$$\begin{aligned} N &\ge e^{k(G(\lambda_j)+\epsilon/2)} \\ &\ge (e^{-\epsilon}X(\lambda_j))^{-k/2}(M(\lambda_j)Y(\lambda_j))^{-\ell/2} \\ &\ge x_{\lambda_j,\delta}^{-k/2}(M(\lambda_j)Y(\lambda_j))^{-\ell/2}. \end{aligned}$$

The balanced-book corollary therefore gives a red K_k or a blue K_ℓ.

Otherwise the average blue degree is greater than $e^{-G'(r)+\delta}(N-1)$, so some vertex has, for all sufficiently large k, at least

$$\exp\{kG(r) + \epsilon k - G'(r) + \delta/2\} \tag{C.10}$$

blue neighbours. Induct on ℓ, starting with the already settled range $\ell \le \rho k$. For $r - 1/k \ge \rho/2$, the mean-value theorem gives

$$k\{G(r) - G(r-1/k)\} = G'(\xi), \qquad G'(r) - G'(\xi) \le k^{-1} \sup_{[\rho/2,1]} |G''| \le \delta/2$$

once k is large. Thus (C.10) is at least the induction threshold $e^{k(G((\ell-1)/k)+\epsilon)}$. Its blue neighbourhood contains a red K_k, or a blue $K_{\ell-1}$ which extends with the chosen vertex. This completes the induction and the uniform ϵ-form of (C.7). $\square$

Proposition C.2 (Rate-certificate correctness)**.** *Assume the repaired book induction in Appendix B and Arb's containment semantics. If the first certificate satisfies (i)–(v), its target obeys the uniform Ramsey-rate bound* (3.3)*. If a finite chain is accepted with the exact handoffs in (v), every target in the chain obeys that bound; in particular the sixth target is F_6.*

Proof. On a closed record interval, (C.2) has strict positive margin in both colour orientations. Together with the already proved uniform prior rate, this allows X, Y_i to be enlarged slightly while retaining both exponent bounds; hence $(X, Y_i) \in \mathcal{R}_*$. Decreasing the first coordinate to $x_{\lambda,\delta}$ preserves interior membership. The finitely many records also give $\sup M_i < 1$, and the verifier proves $F, F' > 0$ and (C.3), which are exactly (C.4)–(C.5).

On $(0, \lambda_*]$, the elementary witness has $Y = 1 - X$. Since $x_{\lambda,\delta} < X$, choose z strictly between them. The elementary recursion places $(z, 1-z)$ in $\mathcal{R}$, while $x_{\lambda,\delta} < z$ and $Y < 1 - z$; therefore the perturbed point lies in $\mathcal{R}_*$. Obligation (iv) proves the remaining analytic hypotheses on the whole open interval, including a positive normalized main slack rather than a sampled extrapolation. Because every target has the form $H(\lambda) + e^{-\lambda} P(\lambda)$ with $P(0) = 0$, its difference from H is $O(\lambda)$. Thus Lemma C.1 proves one accepted link.

For the first link the same elementary argument proves the prior. Exact entrywise target-to-prior equality then permits ordinary induction through the six links, without a rounded or refitted intermediate rate. □

The six accepted certificate identities, in order, are

1. `certificate-higher-order-quintic-v1.json`, `2664bc421cd0cb7489289caa283a3a3f22830580f7c21bf0e6cbe092910bc277`;
2. `certificate-higher-order-quintic-chain-v2.json`, `b5b595b4dcc9d1bdc2b5714f68fef99ec1c566bbd612d35ca19100d173d41c4d`;
3. `certificate-higher-order-sextic-chain-v3.json`, `2052952c3af98074d5442fb736c7a2952146e92051a43fab84e75f099d9e00d7`;
4. `certificate-higher-order-octic-chain-v4.json`, `09253d757516dc258396f82c09a442aaaf9d499640c1cc99b5767e6d51d35942`;
5. `certificate-higher-order-decic-chain-v5.json`, `4ec4dbbe190a08134efe68c3a76ce4b4fb3aabdad230af703c19a28a03ecf9a9`;
6. `certificate-higher-order-tetradecic-chain-v6.json`, `8689e3f80f363ac4b2dada31d6d71d682288eac24ca5e128e22aecfffcd4bde8`.

The canonical replay invokes `verify_chain_arb.py` with `--require-paper-chain`. Before doing any interval work, that mode requires these six filenames in this order, recomputes and compares all six SHA-256 values, and requires the final target tuple to equal the fourteen terminating decimals displayed below, entry by entry.

C.2. **Claim.** Let

$$H(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$$

and interpret every displayed terminating decimal exactly. Define

$$\begin{aligned} P_6(\lambda) = &-0.250000000000000\lambda + 0.019891840059418\lambda^2 \\ &-0.012275954247190\lambda^3 + 0.144110816398083\lambda^4 \\ &+0.006277913420654\lambda^5 - 0.066101623491668\lambda^6 \\ &-0.002287675993602\lambda^7 - 0.058238059505066\lambda^8 \\ &+0.030675864198693\lambda^9 + 0.052472201910597\lambda^{10} \\ &+0.043300454861853\lambda^{11} - 0.042529975074653\lambda^{12} \\ &-0.055263639426635\lambda^{13} + 0.036695886931268\lambda^{14}. \end{aligned}$$

For $F_6(\lambda) = H(\lambda) + P_6(\lambda)e^{-\lambda}$, the exact six-stage chain and the locally closed combinatorial descent prove the local computer-assisted theorem

$$\boxed{R(k,k) \leq (3.78069842779612369875188164975221 62\ldots)^{k+o(k)}\,.}$$

The status is `PROVABLE AS STATED` under the same local proof boundary as the fifth-stage theorem. This is not a published result, an explicit finite-k bound, or a proof-assistant formalization. Independent external human review of this sixth link remains pending.

C.3. **Frozen protocol (declared before target search).** This is a bounded attempt to add **one** descent after the frozen fifth-stage certificate. It will not continue to a seventh stage.

The only admissible prior is the exact ten-entry target tuple in `certificate-higher-order-decic-chain-v5.json`, whose SHA-256 is `4ec4dbbe190a08134efe68c3a76ce4b4fb3aabdad230af703c19a28a03ecf9a9`. No rounded, refitted, or independently retyped prior is admissible. Before search, both existing 256-bit Arb implementations must prove this exact prior strictly concave on all of `(0,1]`.

The construction search is limited to target degrees 10 through 14. A candidate will be frozen only if all of the following predeclared gates hold:

1. its base is at least `0.000001` below the frozen fifth-stage base;
2. dense floating replay plus local scalar refinement finds unbuffered continuous slack at least `0.000150000000000`;
3. the exact certificate has at most 131,072 cells and passes the complete six-stage `verify_chain_arb.py` replay;
4. the separately written `verify_region_direct_arb.py` gives both exact Arb exponent margins at least `0.0000005`;
5. exact `Decimal` comparison proves the P5 target tuple equals the P6 prior tuple entry by entry; and
6. `audit_tests.py` passes unchanged.

A certificate that merely has positive margins but misses one of these gates will be reported as a frontier/FAIL, not as a new Ramsey bound. Floating optimization and interpolation are construction aids only and have no role in the proof.

C.4. **Pre-search prior audit.** The certificate hash was recomputed before optimization and equals the frozen hash above. On its exact target decimals the two independent 256-bit Arb implementations returned

```
verify_arb.prove_prior_concavity:          P5'' <= -0.35642515827888227
verify_region_direct_arb.prove_concavity:  P5'' <= -0.3574394324533535
```

Thus the strict-concavity gate passed before any sixth-stage search. The frozen prior base, computed in Arb rather than binary floating point, is

```
3.780756610409559376317234556232421030227805829166O...
```

C.5. **Bounded degree-10–14 frontier.** The first bounded pass used the exact P5 tuple above, a requested construction margin of `0.00015`, and no target degree outside 10–14. Its diagnostic frontier was

target degree	diagnostic base	sampled minimum slack	disposition
10	3.78075661040956	0.000148091758	no accepted feasible step
11	3.78075661040956	0.000148091758	no accepted feasible step
12	3.78075661040956	0.000148091758	no accepted feasible step
13	3.78069741660317	0.000150109936	frontier only
14, raw	3.78069363690183	0.000150096701	rejected after dense leak

The raw degree-14 point looked best on the construction mesh, but a denser mesh found slack only

```
0.000147957450089331O3 near lambda = 0.3386221294,
```

so it failed the predeclared gate and was not certified. No degree was added and no search range was enlarged. A single guarded degree-14 point was then selected within the already declared range. A broad 20,001-point scan (geometric near the splice and linear thereafter), followed by bounded scalar refinement around every sampled local minimum, returned

```
minimum unbuffered floating slack:
0.00015473769852614172 at lambda ~= 0.33943703594667307

next three refined local minima:
0.00015519556902354736 at lambda ~= 0.9859202132743512
0.00015519607550973370 at lambda ~= 0.6439420227559093
0.00015519870501323751 at lambda ~= 0.8750494757168542
```

```
minimum sampled F6:   0.030260793830868504 at lambda = 0.005
minimum sampled F6’:  0.7658393690544814 at lambda = 1
floating base:        3.780698427796123
```

Thus the unique frozen candidate cleared both the `0.00015` continuous construction floor and the `0.000001` base-improvement floor. The eventual exact Arb base improvement over P5 is

```
0.00005818261343567756535290648O2047549...
```

These floating data only selected the candidate. In particular, they are not used to cover the continuum in the proof.

C.6. **Exact replay and claim boundary.**

C.6.1. *Frozen certificate and exact handoff.* Only the guarded degree-14 target displayed in the Claim was frozen:

```
file: certificate-higher-order-tetradecic-chain-v6.json
SHA-256: 8689e3f80f363ac4b2dada31d6d71d682288eac24ca5e128e22aecfffcd4bde8
cells: 131072
stored left endpoint: 0.0050000000000000001
stored right endpoint: 1
region buffer: 0.0000080000000000000007
constructor predicted worst buffered slack: 0.00011568648667514836
```

The certificate’s ten-entry prior tuple was compared with the P5 target tuple using arbitrary-precision `Decimal`, not binary floating point:

```
P5 target == P6 prior: True
```

Thus the sixth link has no rounded or refitted handoff.

C.6.2. *Complete six-stage 256-bit Arb replay.* `verify_chain_arb.py` replayed the elementary prior and all six exact links and returned

```
PASS: verified 6-stage Ramsey-rate certificate chain
```

The new stage’s strict lower margins and endpoint value were

```
target small slack/lambda:
0.0037785734384423962376319094335140413...

standard Ramsey-region margin:
7.464714995786726e-05 at cell 8

swapped Ramsey-region margin:
5.766703065384951e-05 at cell 131071

large-regime main slack:
4.818471567626107e-05 at cell 131071

F6(1):
1.3299087618219930723187812632036105226...
```

```
exp(F6(1)):
3.7806984277961236987518816497522162752...
```

Every proof-critical margin is strictly positive. The analytic small regime and the 131,072 exact cells cover all $0 < \lambda \le 1$.

C.6.3. *Separately written direct region replay.* `verify_region_direct_arb.py` does not import the primary verifier and directly minimizes both continuous exponent slacks over the complete prior-ratio interval. It returned

```
PASS: independent direct two-sided Ramsey-region replay
prior P5'' worst certified upper:       -0.35743943245335352
worst standard direct exponent slack:    1.0628096485437162e-06 at cell 0
worst swapped direct exponent slack:     5.766703065384951e-05 at cell 131071
```

Both independently replayed margins exceed the predeclared `5e-7` floor. `audit_tests.py` also passed the derivative-sign regression, arbitrary-degree polynomial-derivative regression, and known Horizon union-gap regression.

Consequently every predeclared gate passed and the boxed local asymptotic bound follows. The search stops here: no seventh-stage search was performed.

C.7. **Reproduction commands.** Run from `routes/upper` with the repository environment that supplies `python-flint`:

```
../../.venv/bin/python verify_chain_arb.py --require-paper-chain \
  certificate-higher-order-quintic-v1.json \
  certificate-higher-order-quintic-chain-v2.json \
  certificate-higher-order-sextic-chain-v3.json \
  certificate-higher-order-octic-chain-v4.json \
  certificate-higher-order-decic-chain-v5.json \
  certificate-higher-order-tetradecic-chain-v6.json

../../.venv/bin/python verify_region_direct_arb.py \
  certificate-higher-order-tetradecic-chain-v6.json

../../.venv/bin/python audit_tests.py
```

C.8. **Claim boundary.**

1. The sixth certificate proves one finite descent only; it supplies no convergence theorem for an infinite chain.
2. The inherited asymptotic descent has non-effective $o(k)$ terms, so this is not an explicit finite-k inequality.
3. The numerical proof imports Arb containment semantics through `python-flint` at 256-bit precision.
4. The local combinatorial proof and interval replay have not been formalized in a proof assistant, and this sixth link has not yet received independent external human review or public archival.

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